



REFERENCE ARCHITECTURE

Federated Interconnection Platform for CENTRA Digital

A reference architecture for delivering carrier-neutral, AI-ready interconnect and multi-cloud onramp services across CENTRA's national footprint, anchored by the Charlotte fabric hub.

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1. Executive Summary

CENTRA Digital Interconnect is building the next generation of carrier-neutral, AI-ready interconnection facilities in high-growth, underserved U.S. markets. As CENTRA's footprint expands across Portland, Charlotte, Reno, and Minneapolis, every new site multiplies the operational complexity of stitching carriers, clouds, and customers together. The MaiaEdge platform absorbs that complexity, turns cross-site interconnect into an automated, software-defined service, and converts CENTRA's physical fiber footprint — together with its DIA-only presence in markets like Minneapolis — into a single national, monetizable interconnection fabric.

This reference architecture proposes a federated MaiaEdge deployment across CENTRA's five primary sites — Charlotte, Portland (ME), Reno-1, Reno-2, and Minneapolis. Charlotte is designated as the External Fabric Hub, landing the Equinix Fabric handoff and providing on-demand cloud onramps to AWS, Azure, GCP, and SaaS / partner providers. Dedicated long-haul wavelengths connect Portland, Reno-1, and Reno-2 to Charlotte for premium, deterministic transport. Minneapolis demonstrates a key property of the architecture: even without dedicated fiber, a site can join the fabric and reach every service through the redundant DIA-1 / DIA-2 mesh. MaiaEdge Path Border Controllers (PBCs) and Port Extenders at each site terminate carrier and customer fibers, replace the static Meet-Me Room (MMR) with a software-driven cross-connect plane, and stitch encrypted Layer-2 paths across every site, regardless of whether it has waves or only DIA.

Why this matters for CENTRA

Every CENTRA site becomes a programmable interconnection node — whether it is wave-connected (Charlotte, Portland, Reno-1, Reno-2) or DIA-only (Minneapolis). New customers and partners self-provision cross-connects in minutes; carriers, hyperscalers, and content providers attach to a single fabric; CENTRA and its tenants see every service end-to-end, path-by-path and hop-by-hop; and CENTRA earns recurring revenue on every virtual port and cloud onramp sold through the MaiaEdge Marketplace.

Strategic Outcomes at a Glance

Business Outcome	How the Architecture Delivers It
Faster time-to-revenue per site	Software-driven cross-connects let CENTRA light up a new tenant or carrier in minutes, not weeks of MMR cabling.
Differentiated AI / hyperscaler offer	Encrypted L2 paths across diverse DIAs and a Charlotte Equinix Fabric hub deliver low-latency, multi-cloud onramp from any CENTRA site.
Higher revenue per cabinet	MaiaEdge Marketplace converts physical ports into recurring virtual port, cross-connect, and cloud onramp revenue streams.



Operational leverage	One unified control plane replaces site-by-site manual processes, letting a small NOC team operate the full national footprint.
Coverage without fiber lock-in	DIA mesh extends the fabric to sites such as Minneapolis where dedicated fiber waves are not yet available — no need to wait for new builds.
End-to-end service visibility	Every tenant service is observable path-by-path and hop-by-hop across the entire fabric — turning a chronic colo-customer pain into a differentiated CENTRA value-add.
Carrier, cloud, and content (C3) magnetism	Federated fabric makes CENTRA an attractive landing point for partners; every new partner increases the value of the marketplace flywheel.



2. About CENTRA Digital Interconnect

CENTRA Digital Interconnect (centradigital.com) is a Washington, DC-based digital infrastructure company that designs, builds, and operates carrier-neutral interconnection data centers in high-demand, high-growth U.S. markets. Backed by a \$230M investment led by Columbia Capital, CENTRA is purpose-built around the Carrier, Cloud, and Content (C3) ecosystem — the three customer segments whose interconnect needs are accelerating fastest in the AI era.

CENTRA Footprint Referenced in this Architecture

Site	Role in the v5 Architecture	Connectivity Profile
Charlotte, NC	External Fabric Hub. Equinix Fabric on-ramp landing site.	Carrier-neutral 2N / N+1 facility, Carolinas MMR. Wave terminus for Portland, Reno-1, and Reno-2. Equinix Fabric uplink.
Portland, ME (PWM1 / 340 Cumberland)	Northeast interconnection node. Wave to Charlotte.	Maine's only carrier-neutral telecom facility. Long-haul fiber from Consolidated, Crown Castle, FirstLight, Spectrum, Verizon. NNENIX home.
Reno-1 (Nevada)	West-coast interconnection node. Wave to Charlotte.	Existing Reno presence; pairs with Reno-2 for in-market diversity.
Reno-2 (RNO2)	West-coast interconnection node. Wave to Charlotte. AI-ready power profile.	92,000 sq ft, 2 MW launch / 7.5 MW at full buildout. Reno's first purpose-built interconnection data center.
Minneapolis (MSP1)	Upper-Midwest interconnection node. DIA-only — no dedicated wave.	150,000 sq ft, 10+ MW, Tier III+. Reaches the fabric and Equinix hub via redundant DIA-1 / DIA-2 mesh — proving the architecture works without fiber.

Strategic Drivers

This architecture is shaped by three observable realities in CENTRA's market:

- AI workloads are reshaping interconnect demand. Training and inference traffic spans GPU clusters, object stores, and multi-cloud tenants — all of which require deterministic, low-latency, and encrypted paths between sites and clouds.
- Carrier-neutrality is CENTRA's competitive moat. The architecture must preserve and amplify CENTRA's neutral position by giving every carrier, cloud, and content provider equal, programmatic access to every CENTRA site.



- Marketplaces win the C3 ecosystem. Sites that turn physical interconnect into an on-demand, self-service product capture more share of the C3 wallet than sites that rely on manual MMR provisioning. A site without dedicated fiber must still be able to participate — this architecture ensures it can.



3. Reference Architecture Overview

At a high level, the architecture deploys a federated MaiaEdge fabric across all five CENTRA sites. Each site contains a redundant pair of MaiaEdge Path Border Controllers (PBCs) and Port Extenders that terminate customer fibers, carrier fibers, and CENTRA's internal cabinets. PBCs across sites are stitched together over two diverse Dedicated Internet Access (DIA) meshes — DIA-1 and DIA-2 — each running on independent fiber paths and providers.

Charlotte is designated the External Fabric Hub. It carries the wave into the Equinix Fabric, which in turn provides on-demand cloud onramp into AWS, Azure, GCP, and a broad set of SaaS and partner providers. Three additional long-haul waves — Portland ↔ Charlotte, Reno-1 ↔ Charlotte, and Reno-2 ↔ Charlotte — give wave-connected sites a deterministic path to the hub and to the cloud onramp. Minneapolis joins the same fabric without a dedicated wave, using the DIA-1 / DIA-2 mesh to reach Charlotte and every cloud onramp the hub exposes. Every L2 path between PBCs is encrypted and authenticated at line rate, regardless of whether it traverses a dedicated wave or the DIA underlay.

Architecture Diagram

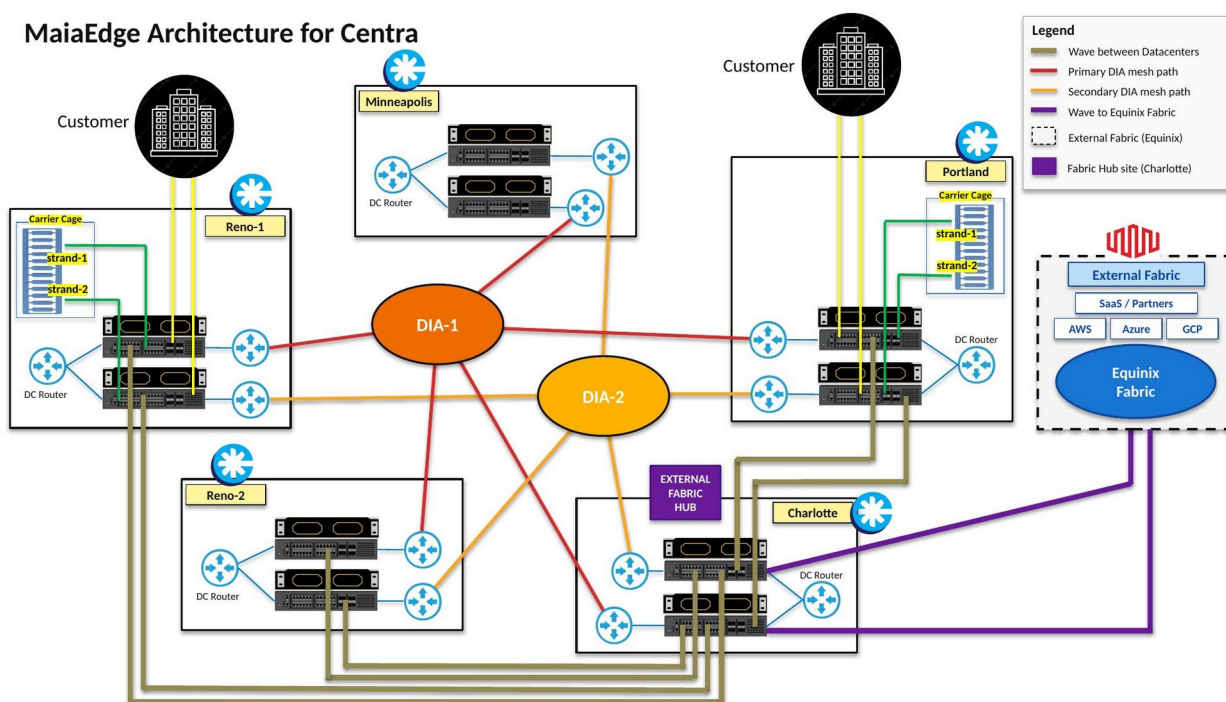


Figure 1. MaiaEdge federated architecture across CENTRA's five primary sites. Charlotte hosts the External Fabric Hub; waves connect Portland, Reno-1, and Reno-2 to Charlotte; Minneapolis reaches the fabric via DIA mesh.

Reading the Diagram

Element	Meaning
Solid red line	Primary DIA mesh path (DIA-1) between PBCs.



Solid yellow / orange line	Secondary DIA mesh path (DIA-2) — physically and provider-diverse from DIA-1.
Purple line	Wave from Charlotte into the Equinix Fabric (cloud onramp).
Olive / khaki lines	Waves between datacenters: Portland ↔ Charlotte, Reno-1 ↔ Charlotte, Reno-2 ↔ Charlotte.
Green strands	Carrier fiber strands terminating into the carrier cage and PBC.
Yellow strands	Customer fiber strands terminating into the carrier cage and PBC.
Purple box near Charlotte	Hub site — designated External Fabric Hub.
Black dashed box on right	External Fabric (Equinix Fabric, AWS, Azure, GCP, SaaS / Partners).
Minneapolis (no waves)	Fabric participation via DIA mesh only — no dedicated fiber required to reach the hub.



4. Core Components

4.1 MaiaEdge Path Border Controller (PBC)

The Path Border Controller is the cornerstone of the architecture. It is a 1RU appliance that combines Layer-2 and Layer-3 functions, terminates carrier and customer fibers, and establishes encrypted, authenticated paths to every other PBC in the fabric. PBCs are deployed in redundant pairs at every CENTRA site so that any single device, line card, or upstream fiber failure is absorbed without customer impact.

- Dual 100 Gbps uplink ports per PBC for carrier and DIA-mesh connectivity.
- Line-rate encryption and authentication on every L2 path; no separate IPsec concentrator or third-party security overlay required.
- Centralized control plane via the MaiaEdge Path Computation Engine (PCE) — no per-site BGP, VLAN, or stitching logic to maintain.
- Deploys at the network boundary — carrier hotels, MMRs, and PoPs — exactly matching CENTRA's site topology.

4.2 MaiaEdge Port Extender

The Port Extender is a companion switch that scales tenant port density on each PBC pair. Each Port Extender adds up to 48 additional tenant-facing ports under the same control plane and security model as the PBC. This lets CENTRA grow customer density inside a cabinet without re-architecting the underlying fabric or introducing a second management layer.

4.3 DIA Mesh (DIA-1 / DIA-2)

Inter-site connectivity uses two diverse Dedicated Internet Access (DIA) circuits as the underlay. DIA-1 and DIA-2 are deliberately sourced from different upstream providers and routed over physically diverse fiber paths wherever feasible. MaiaEdge PBCs build encrypted L2 tunnels over both DIA meshes simultaneously, so a failure of any single carrier, fiber path, or DIA provider triggers automatic re-routing without human intervention. The DIA mesh is the workhorse that lets Minneapolis fully participate in the fabric today, before any dedicated wave is in place.

4.4 Long-Haul Wave Services

For premium, deterministic transport between primary sites, CENTRA supplements the DIA mesh with dedicated wavelengths terminating in Charlotte: Portland ↔ Charlotte, Reno-1 ↔ Charlotte, and Reno-2 ↔ Charlotte. Waves provide a SLA-grade, low-jitter underlay for high-bandwidth tenant flows and are exposed to customers as a premium service tier in the MaiaEdge Marketplace. Sites without a wave (Minneapolis today) operate on DIA mesh only; CENTRA may add dedicated waves as those markets justify the build.



4.5 Carrier Cage and MMR Modernization

The carrier cage at each site terminates incoming carrier fibers (green strands) and customer fibers (yellow strands) directly into the MaiaEdge PBCs. The PBCs effectively serve as a software-defined Meet-Me Room: any carrier port can be stitched to any customer port — locally or across sites — through the MaiaEdge control plane, replacing manual cross-connect tickets with API- and portal-driven provisioning.

4.6 Equinix Fabric Hub at Charlotte

Charlotte is the External Fabric Hub. It carries the wave into the Equinix Fabric, which in turn provides on-demand cloud onramp into AWS, Azure, GCP, and a broad set of SaaS and partner providers. Because the Equinix Fabric is reached through MaiaEdge, every other CENTRA site — Portland, Reno-1, Reno-2, and Minneapolis — inherits the same multi-cloud onramp without needing its own physical Fabric port. Minneapolis specifically reaches this onramp over the DIA mesh, demonstrating that fabric participation is not gated by dedicated long-haul fiber.

4.7 MaiaEdge Marketplace

The Marketplace is the commercial expression of the fabric. CENTRA — and its partner providers — can advertise and sell products such as cross-site L2 connectivity, cloud onramps, partner-to-partner peering, and value-added services through a unified portal and API. Provisioning is automated end-to-end: a customer order results in encrypted L2 paths and PBC port assignments without any change-control ticket or truck roll, regardless of whether the underlay is a dedicated wave or the DIA mesh.



5. Service Catalog Enabled by the Architecture

The architecture is intentionally designed to expose services that match the buying patterns of carrier, cloud, and content (C3) customers, and to make those services orderable through both portal and API.

Service	What CENTRA Sells	Underlying Architecture
Cross-site Private L2	Encrypted Layer-2 circuits between any two CENTRA sites at customer-selectable bandwidths.	MaiaEdge PBC encrypted L2 paths over the DIA-1 / DIA-2 mesh, augmented by wave services where available.
Multi-Cloud Onramp	On-demand virtual connections from any CENTRA site (including DIA-only Minneapolis) to AWS, Azure, GCP, and supported SaaS providers.	Equinix Fabric hub at Charlotte; cross-site stitching via MaiaEdge.
Programmable Cross-Connect (Virtual MMR)	Self-service cross-connects between any two ports inside a CENTRA site, provisioned in minutes.	PBC + Port Extender as software-defined MMR.
Premium Wave Services	Dedicated long-haul wavelengths for high-bandwidth, deterministic flows into the Charlotte hub.	Portland ↔ Charlotte, Reno-1 ↔ Charlotte, Reno-2 ↔ Charlotte waves; complement DIA mesh for premium tiers.
Partner-to-Partner Peering	Federated peering between CENTRA-hosted partners and partners hosted in other MaiaEdge-enabled providers.	MaiaEdge federated control plane; trust and policy enforced per partner.
AI / GPU Interconnect	Low-latency, encrypted paths between AI compute, storage, and inference endpoints across sites.	DIA mesh + line-rate encryption on PBCs; waves where available; MSP1 and RNO2 provide AI-ready power and density.



6. Resiliency and Redundancy Model

Resiliency is engineered at every layer so that no single failure removes a customer service. The principles below are reflected directly in the diagram.

Failure Domain	Protection Mechanism
PBC device failure	Redundant PBC pair at every site; live failover with no session loss.
Port Extender failure	Tenants can be dual-homed across two Port Extenders within the cabinet.
Carrier fiber cut (single strand)	Strand-1 / Strand-2 termination per cabinet; PBC re-homes tenant traffic.
Single DIA provider outage	DIA-1 and DIA-2 meshes are provider- and path-diverse; PBCs continuously use both. Critical for Minneapolis, which relies on DIA only.
Wave outage (Portland / Reno-1 / Reno-2 ↔ Charlotte)	Affected site's traffic falls back to the DIA-1 / DIA-2 mesh automatically; service continues at DIA performance until the wave is restored.
Site outage at non-hub site	DIA mesh routes traffic around the impacted site; remaining sites continue to reach the Charlotte hub and Equinix Fabric.
Hub site (Charlotte) impairment	Local services unaffected at other sites; cloud-onramp failover plan activated. Long-term roadmap: a second External Fabric Hub for full hub redundancy.

7. Security and Trust Model

Security is built into the fabric itself rather than bolted on. Every L2 path between PBCs is encrypted and authenticated at line rate — protecting tenant traffic across the DIA mesh and waves alike, without requiring per-tenant VPN concentrators or out-of-band crypto appliances. Tenant isolation, policy enforcement, and audit are managed centrally.

- Encryption: Line-rate encryption on every inter-PBC L2 path. Keys are rotated and managed centrally by the MaiaEdge control plane.
- Authentication: Every PBC mutually authenticates with its peers; rogue or swapped devices cannot join the fabric.
- Tenant isolation: Each customer is logically isolated end-to-end; cross-tenant leakage is prevented at the PBC silicon, not by post-hoc ACLs.
- Federated trust: Partner-to-partner peering uses explicit, auditable trust policies — CENTRA controls who can advertise into its marketplace and what they may see.



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- Operational separation: CENTRA management traffic is logically separated from tenant data planes on every PBC.



8. End-to-End Service Visibility

One of the most chronic, unsolved problems in the colocation industry is what happens when a tenant says "my service is slow." The customer can see only their own end. The colo provider can see only its own facility and underlay. Carriers can see only the segments they operate. The result is finger-pointing, long mean-time-to-resolution, and a recurring trust tax on the colo-customer relationship.

The MaiaEdge architecture eliminates this opacity. Because every L2 path between sites is constructed and managed by the MaiaEdge control plane, the fabric itself produces continuous, path-aware telemetry — and CENTRA and its tenants can see exactly how each service is performing, path-by-path and hop-by-hop, end-to-end.

8.1 What CENTRA and its Customers Can See

- Per-service path view: for every tenant service, the exact path through the fabric — from the ingress PBC port, across the chosen underlay (wave or DIA), to the egress PBC port at the destination site or cloud onramp.
- Hop-by-hop telemetry: latency, jitter, loss, and utilization measured at every PBC hop along the path, with deltas between hops to localize impairments.
- Underlay correlation: when a service degrades, CENTRA can immediately tell whether the cause is the wave, a specific DIA provider, a carrier strand, or a tenant-side configuration — no triage call required.
- Per-tenant SLA visibility: tenants see their own service performance against the contracted tier (premium wave vs. best-effort DIA), exposed through the Marketplace portal and API.
- Historical and real-time: streaming telemetry for live troubleshooting plus rolled-up history for capacity planning, SLA reporting, and post-incident review.

8.2 Why This is a CENTRA Differentiator

Industry Pain Point	How the Architecture Resolves It
Customer reports degradation; cause is unknown	Path view + hop-by-hop telemetry localize the impairment in seconds, before a ticket even opens.
Finger-pointing between colo, carrier, and customer	Single source of truth across every hop — including DIA underlay segments — removes ambiguity about who owns the issue.
Tenants have no visibility once traffic leaves their cabinet	Tenants see their own service performance end-to-end through the Marketplace portal.
Long MTTR on cross-site or cloud-onramp issues	Operators can pinpoint the offending hop or wave/DIA segment immediately and re-route via the alternate underlay.
SLA disputes are hard to resolve	Per-service, per-tier telemetry provides the evidence base for SLA credits or premium-tier upsell conversations.



DIA-only sites (e.g., Minneapolis) feel "second class"	Same end-to-end visibility applies to DIA paths — tenants on DIA-only sites get the same observability story as wave-connected sites.
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8.3 How It is Delivered

- PBCs emit streaming telemetry on every L2 path they participate in — both wave and DIA underlays — to the MaiaEdge Path Computation Engine (PCE).
- PCE correlates per-hop telemetry into per-service path views and exposes them through portal, API, and webhooks for CENTRA's CRM, ticketing, and observability stacks.
- Tenant-scoped views are filtered automatically: each customer sees only the paths and metrics that belong to its services.
- Alarms and anomaly detection are wired into CENTRA's NOC tooling, with rich context (path, hops, underlay, tenant) included in every alert.



9. Operational Model

The architecture is designed so that one small operations team can run the full national fabric. Day-to-day operations shift from hands-on cabling and ticket-driven cross-connects to API-driven service orchestration.

9.1 Day-0: Site Bring-up

- Install redundant PBC pair and Port Extenders in each new CENTRA cabinet.
- Terminate carrier and customer fibers from the carrier cage into the PBCs.
- Provision DIA-1 and DIA-2 circuits with diverse providers; bring up encrypted L2 paths. (For wave-connected sites, also bring up the wave to Charlotte.)
- Register the new site in the MaiaEdge control plane and Marketplace — sites without a wave are activated identically.

9.2 Day-1: Customer / Carrier Onboarding

- Customer or carrier port assigned via the MaiaEdge portal or API.
- Cross-connect, L2 path, or cloud onramp ordered — provisioning completes in minutes, not weeks.
- Billing meter starts automatically against the ordered service tier.

9.3 Day-2: Steady-state Operations

- Single pane of glass for fabric health, capacity, and tenant SLAs across wave and DIA underlays.
- Streaming telemetry from every PBC into CENTRA's observability stack.
- Automated change windows; no manual MMR ticket queue.
- API-first integration with CENTRA's CRM, ticketing, and billing systems.

10. Business Outcomes and Monetization Model

The architecture is engineered to convert physical fiber, wave, and DIA investments into a recurring, software-driven revenue model. The five outcomes below are the primary business cases for the deployment.

Outcome	Indicative Lever	Why the Architecture Enables It
Higher revenue per port	Convert physical ports into multiple billable virtual services.	PBC + Port Extender expose each physical port as many software-defined services that can be sold, metered, and renewed independently.
Faster time-to-revenue	Days-to-minutes provisioning for new tenants.	Automated cross-connect and cloud-onramp eliminate ticket queues and truck rolls.
Lower operating cost per site	One unified control plane vs. site-specific tooling.	Federated MaiaEdge management replaces site-by-site MMR processes.



New revenue: cloud onramp	Sell AWS / Azure / GCP onramp from every CENTRA site.	Equinix Fabric hub at Charlotte is reachable from every site via wave or DIA — including Minneapolis with no dedicated fiber.
Two-tier transport pricing	Premium wave tier and best-effort DIA tier for the same logical service.	Same MaiaEdge L2 service offered over wave (premium SLA) or DIA (best-effort) — same provisioning, different commercial tier.
Visibility-as-a-product	Sell a paid "observability" tier (or include it in premium) so tenants can see their own service end-to-end.	Per-service, hop-by-hop telemetry from every PBC; tenant-scoped views in the Marketplace portal.
Lower MTTR and ticket volume	Cut time-to-resolve cross-site / onramp issues and reduce inbound tickets per tenant.	Path-aware telemetry localizes impairments before tickets open and removes finger-pointing across colo / carrier / customer.
Marketplace flywheel	Each new partner increases value of every existing port.	Federated control plane lets partners advertise into and consume from the MaiaEdge Marketplace.

11. Suggested Roadmap

The deployment is naturally phased so CENTRA earns revenue and operational experience as the footprint grows.

Phase	Scope	Outcome
Phase 0 — Pilot	Deploy redundant PBCs at Charlotte and one second site (e.g., Portland). Bring up DIA-1 / DIA-2 mesh and the Portland ↔ Charlotte wave. Stand up the Equinix Fabric hub at Charlotte.	Validate L2 path performance, encryption, and self-service provisioning. First marketplace transactions on a pilot tenant set.
Phase 1 — National Mesh	Extend PBCs to Reno-1, Reno-2, and Minneapolis. Activate full DIA mesh. Light Reno-1 ↔ Charlotte and Reno-2 ↔ Charlotte waves.	Every CENTRA site is interconnect-ready and Marketplace-enabled. Minneapolis is live on DIA-only, demonstrating the fabric extends beyond the wave footprint.
Phase 2 — Marketplace Scale-out	Onboard initial wave of carrier, cloud, and content partners. Publish service catalog. Integrate with CENTRA CRM and billing.	First recurring marketplace revenue; partner-driven port density growth.



Phase 3 — Federation	Federate the MaiaEdge fabric with partner providers; expose CENTRA interconnect in third-party marketplaces and vice versa.	CENTRA reach extends beyond owned sites; flywheel acceleration.
Phase 4 — Wave Densification & Second Hub	Add Minneapolis wave (when justified) and a second External Fabric Hub for hub redundancy and East/West cloud-onramp diversity.	Premium-tier coverage matches the entire footprint; full hub-level HA.

12. Key Risks and Mitigations

Risk	Mitigation
Hub-site dependency on Charlotte for cloud onramp	Roadmap includes a second External Fabric Hub; near-term mitigation via Equinix Fabric route diversity within the hub.
DIA provider concentration — especially impactful for Minneapolis	Two diverse providers from day one; quarterly review of carrier diversity and route-disjoint testing. Minneapolis is the canary site for DIA-only performance assurance.
Wave-tier customer expectations for non-wave sites	Service catalog clearly differentiates wave (premium SLA) from DIA (best-effort) tiers; Minneapolis customers buy DIA tier explicitly.
Operational change for CENTRA staff (manual MMR → API)	Phased rollout with MaiaEdge enablement and shadow-running existing MMR process during pilot.
Marketplace cold-start (few initial partners)	Anchor on existing CENTRA carrier and cloud partners in Phase 2; use federation in Phase 3 to import demand.
Encryption / compliance concerns from regulated tenants	Line-rate L2 encryption + auditable tenant isolation; mapping to common compliance frameworks documented as part of onboarding.

13. Glossary

Term	Definition
PBC	Path Border Controller. MaiaEdge's 1RU edge appliance combining L2/L3 and line-rate encryption.
PCE	Path Computation Engine. The cloud-based MaiaEdge control plane that computes



	optimal paths and enforces SLAs across PBCs.
Port Extender	Companion switch that adds up to 48 tenant-facing ports under the PBC control plane.
DIA	Dedicated Internet Access — high-capacity carrier circuits used as the underlay for the MaiaEdge mesh. The sole underlay for sites without dedicated fiber, such as Minneapolis.
Wave	Dedicated long-haul wavelength service. In v5: Portland ↔ Charlotte, Reno-1 ↔ Charlotte, Reno-2 ↔ Charlotte, plus Charlotte ↔ Equinix Fabric.
MMR	Meet-Me Room — the colocation room where carriers and tenants traditionally cross-connect; here, replaced by software.
Equinix Fabric	Equinix's on-demand interconnection platform for cloud onramps and partner peering.
Hub Site	Designated site (Charlotte) that lands external fabric services for the rest of the footprint.
C3	Carrier, Cloud, Content — CENTRA's primary customer ecosystem.
Marketplace	MaiaEdge's commercial layer for advertising, ordering, and provisioning fabric services.

14. Recommended Next Steps

- Confirm the in-scope sites and fiber inventories at each location with CENTRA operations.
- Validate diverse DIA provider candidates per site — with extra rigor for Minneapolis given its DIA-only profile.
- Agree on the Phase-0 pilot scope (Charlotte + Portland), success criteria, and target go-live.
- Align Marketplace integration on CRM, ticketing, and billing endpoints.
- Schedule joint operational readiness review prior to Phase-1 rollout.

— End of document —